Growth Stage-Specific Climate Features Substantially Improve Wheat Yield Prediction in North India

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***Abstract*:** Historically, crop-climate modeling has been done by seasonally aggregating climate variables e.g. by averaging temperature and rainfall throughout the entire crop growth period. Such strategy has been demonstrated to describe 25-35 percent of global yield fluctuation (Ray et al., 2015). Nevertheless, these models presuppose identical crop response over the stages of growth and this is physiologically incorrect. Wheat has a stage-sensitive stress to climate with flowering being only sensitive to temperatures not exceeding 31°C and vegetative growth can withstand substantially higher temperatures (Porter and Gawith, 1999). This research investigates the question on whether any temporal disaggregation of climatic characteristics by growth stages enhances predictability of yields over conventional climatic extension of seasons. We used parallel regression modeling on 13 year long daily climate observations of North India (2010-2022) (n = 169 district-years) with Model A fitted on seasonal aggregation and Model B fitted with growth stage-specific characteristics. There are significant increases of predictive performance with Model B (R^2 = 0.199 vs 0.040; F(9,156) = 3.43, p = 0.0007) being much better than Model A. The number of features that were statistically significant in the growth stage (7) was 7 times more than the number of features that were statistically significant in the seasonal feature (one). The predictor with the highest predictive ability (r = 0.206, p = 0.007), was the extreme heat at maturation stage, then the grain-filling temperature (r = 0.186, p = 0.015), and flowering heat stress (r = 0.168, p = 0.029), and maturation rainfall had a negative relationship (r = -0.193, p = 0.012). The crisis of stage-specific analysis in growth stages was dramatically demonstrated by the heatwave in March 2022 in North India where extreme heat during flowering and grain filling caused widespread yield losses of 10-15% in spite of the seasonal averages remaining within the normal range. This experience proves that traditional seasonal forecasting may not be effective in relation to climate effects where extreme weather events coincide with sensitive development periods.

Keywords - Wheat Yield, Temporal Disaggregation, Growth Stage Sensitivity, Climate Aggregation, Phenological Specificity, North India.

I. Introduction

Being one of the primary staple crops, wheat (Triticum aestivum L.) contributes to almost a fifth of the world food supply. Nonetheless, there is an increasing threat of climate change to the production of wheat. (Asseng et al. 2015) estimate the future world yield of wheat to decline by about 6 percent with each 1°C of rise in the temperature. The practicality of knowledge of stage-specific climate effects was clearly illustrated in the March 2022 heatwave in North India. Higher than ever temperatures of over 40°C in late February and March when critical flowering and grain filling were underway in the wheat crop led to large-scale losses of 10-15% in Punjab, Haryana and Uttar Pradesh. The most valuable lesson that this event taught about climate-yield models was that seasonal averages in climate in the 2021-22 growing season were not out of historical performance. The average temperature during the whole period of November-April did not contain any extraordinary deviation. Yet yields fell sharply. The agricultural sustainability is also facing challenges in North India that produces almost 85 percent of all wheat in the country and even experiences higher temperatures and inconsistent rainfall (Jain et al., 2019). Our systematic study was inspired by this practical failure of the traditional seasonal methods. The seasonal aggregation models that mean climate variables over the whole growing period were not able to predict the extent of 2022 effects since it hid an important fact: extreme heat happened at the most vulnerable stages of development. Instead of averaging climate data based on seasons, it should have been based on growth stages and the high risk during flowering would have become obvious at once, perhaps allowing early warning systems and adaptive responses. Majority of the crop-climate relationships modeling is based on an aggregation over seasons, with the temperature and precipitation averaged over the entire growing season. This strategy demonstrates that climate variation accounts for 25-35% of the global yield variations (Ray et al., 2015). But seasonal averaging is that crops behave in the same way throughout all growth stages which is not the case with physiological evidence. Wheat developmental stages react differently to the changes in temperatures. There are examples of 31°C of maximum temperature that will be withstood during flowering but temperatures that are much higher are with standable during vegetative growth (Porter and Gawith, 1999). Pollen sterility and irreversible loss of yield may be triggered by the impact of heat stress on the short flowering period (10-15 days) (Prasad & Djanaguiraman, 2014).

Even though the shortcomings of seasonal averaging are generally admitted (Lobell et al., 2011), the method is still used because of the lack of available data and the convenience of the analysis. This generates a gap in methodology. Physiological studies have established well climate thresholds in different developmental stages but empirical models still lump climate variables together across these heterogeneous stages of development. Previous research has not compared growth stage-specific characteristics to seasonal aggregation using controlled and same datasets. The gap is filled in this study using parallel modeling, using the same data on North Indian wheat (2010-2022, n = 169). We examine whether climate is better predicted by dividing the climate data by stage of growth and to determine times of critical vulnerability. We find that the stage of growth aspects are found to contribute substantially to explained variance (R2 = 0.199 vs. 0.040 F(9,156) = 3.43, p < 0.001) (392 percent higher). Seven features of growth stage were found to be statistically significant as opposed to one seasonal feature. The major predictors are high temperatures of maturation (r = 0.206, p = 0.007), temperature of grain filling (r = 0.186, p = 0.015) and temperature of flowering stress (r = 0.168, p = 0.029).

These findings suggest that seasonal averaging hides significant climate-yield relationships which are clear when data are disaggregated by growth stage. The method aids in discovering the crucial moments of weakness and facilitates the creation of specific climate adaptation measures.

LITERATURE REVIEW

2.1 Evolution of Crop–Climate Modeling

Over the past two decades, statistical models—particularly regression analysis—have become central to understanding the relationship between climate and crop production. These models typically link seasonal averages of temperature and rainfall with end-of-season yields (Lobell & Field, 2007; Schlenker & Roberts, 2009; Auffhammer et al., 2012). Their widespread adoption is largely due to their simplicity and ability to explain yield variability across regions and time.

Empirical evidence demonstrates their effectiveness. For instance, (Ray et al., 2015) showed that seasonal climate variability explains nearly one-third of global wheat yield variation. Similarly, (Zhao et al. 2017) confirmed across multiple statistical models that increased seasonal temperatures consistently reduce wheat yields. Regional studies in North India further indicate that seasonal climate variables account for 25–35% of yield variability (Auffhammer et al., 2012).

Despite these successes, these models rely heavily on seasonal aggregation, combining all weather data between planting and harvest into a single average. While some studies acknowledge that crop responses may vary across growth stages, this variability is rarely explored in depth. Consequently, the assumption that wheat responds uniformly to climatic conditions throughout its lifecycle remains a key limitation.

2.1 Insights from Plant Science on Wheat Sensitivity

Plant physiology research clearly shows that wheat exhibits stage-specific sensitivity to climatic conditions, challenging the validity of seasonal averaging. The most critical phase is flowering (anthesis), during which temperatures above 31°C can damage pollen viability, significantly reducing grain formation (Porter and Gawith, 1999). This phase lasts only 10– 15 days, yet determines final grain number. Even short heat events (e.g., 35°C for a few hours) can cause irreversible yield loss (Prasad & Djanaguiraman, 2014). The grain filling stage represents another crucial period, where grain weight is determined. Moderate temperatures (25–28°C) are optimal, but temperatures above 30°C accelerate maturation, shortening the grain-filling duration and resulting in smaller grains (Stone & Nicolas, 1995). Unlike flowering, this stage allows limited recovery from stress.

In contrast, early growth and maturation stages are relatively resilient. Wheat can tolerate a wide temperature range (15– 30°C) during vegetative growth, and grain hardening remains stable until temperatures exceed 35°C (Porter and Gawith, 1999). This variability highlights a critical flaw in averaging approaches: identical temperatures can have drastically different impacts depending on the growth stage.

This variability highlights a critical flaw in averaging approaches: identical temperatures can have drastically different impacts depending on growth stage—a principle validated by the differential impacts observed across the 2022 growing season.

2.2 Research Gap: Recognition Without Resolution

This highlights a critical limitation in existing approaches. The problem has been known for several years, but little has been done to address it. (Lobell et al., 2011) went as far as stating, ***"Seasonal averaging may obscure critical periods,"*** and then proceeded to use seasonal averaging anyway.

When (Tack et al. 2015) examined American wheat, they recognized that different growth stages are likely to respond differently to weather conditions, yet still adopted the seasonal approach. The gap in research arises from a simple reason: seasonal analysis is easier to implement. The persistence of this limitation is largely due to data constraints and methodological convenience, as seasonal aggregation simplifies analysis. However, "easy" is not always "optimal," especially since precise temperature thresholds for each growth stage have been known for decades.

What remains missing is a direct comparison. Has any study compared seasonal averaging and stage-specific analysis using the same dataset? Previous attempts to incorporate finer temporal resolution have been limited in one of two ways: either different datasets were used for different approaches, making it difficult to isolate the cause of differences, or multiple growth stages were aggregated into broad categories such as ***"reproductive period"*** (Gourdji et al., 2013), which still obscures critical variability.

Most existing research has focused on developed regions with dense weather monitoring networks, leaving climate-vulnerable regions such as South Asia relatively understudied. This study addresses these gaps by applying both approaches to identical data from North India, enabling precise measurement of the information lost when climate variables are averaged across growth stages rather than analyzed individually.

The March 2022 heatwave provided a natural experiment demonstrating the real-world consequences of this research gap. Seasonal models failed to anticipate yield impacts because they averaged extreme late-season heat across the entire growing period, obscuring the temporal coincidence of
heat stress and vulnerable developmental stages.

METHODOLOGY

3.1 Study Area and Data Sources

North India was selected for this study because it represents one of the world's most critical wheat-producing regions, accounting for 85% of India's total wheat output. The region faces mounting climatic stresses such as rise in temperatures, the number of heat waves, and unpredictable rainfall distribution. Such properties allow it to serve as a good venue in assessing the influence of temporal resolution in climate data on the accuracy of yield prediction. In order to represent the agro-climatic diversity, thirteen districts were chosen in four different states where wheat was the most dominant Rabi (winter) crop (Table 1). Districts chosen cover extensive climatic range, semi-arid areas: in Rajasthan; humid subtropical areas: in Punjab and Haryana, and sub-Himalayan areas: in western Uttar Pradesh. This geographic variation increases representativeness of results of the findings in various growing conditions.

TABLE 1: STUDY DISTRICT CHARCTERISTICS

District	State	Coordinates	Agro-Climate	Mean Yield (q/ha)	Primary Constraint
Ludhiana	Punjab	30.9°N, 75.9°E	Humid subtropical	49.1	Late-season heat
Bathinda	Punjab	30.2°N, 74.9°E	Semi-arid	49.8	Water stress
Karnal	Haryana	29.7°N, 77.0°E	Humid subtropical	48.8	Terminal heat
Saharanpur	Uttar Pradesh	30.0°N, 77.6°E	Sub-Himalayan	36.1	Fog, cold stress
Meerut	Uttar Pradesh	29.0°N, 77.7°E	Humid	42.3	Heat, rainfall variability
Sri Ganganagar	Rajasthan	29.9°N, 73.9°E	Semi-arid	41.4	Drought, extreme heat

3.2 Stage of Growth and Temporal Disaggregation

The wheat crop in North India lasts, on average around 150 days whereby the sowing period is in the middle of the month of November, and the harvest period is in the middle of the month of April. Instead of categorizing this whole time interval identically, we categorized it into physiologically distinct growth phases on the basis of set phenological calendars of the area and also on the basis of physiological studies of wheat development. There are three important development milestones, which we defined: flowering (February 1-15), grain filling (February 16 – March 15) and maturation (March 16 – April 10). These dates are the average date of varieties that have medium duration (120- 140 days to maturity) that are typical of plants that are generally cultivated in our study areas. The flowering or anthesis is the short period of time during which the pollen is emitted and fertilization takes place, which defines the number of grains that are going to be produced. It is period of grain filling when fertilized grains grow starch and protein that defines the final weight of grains. The last stage is maturation where grains become hard and dry awaiting harvest. We acknowledge that actual phenology varies with planting dates, varieties, and yearly weather. However, fixed calendar dates provide methodological consistency essential for controlled comparison, mirroring how seasonal aggregation standardizes periods. In order to make comparisons, we also determined a full seasonal period to be between November 1 and April 30 inclusive in terms of the full wheat cycle including pre-sowing soil moisture conditions and harvest. This six-month period is the conventional method of seasonal aggregation research

3.3 Climate Feature Engineering

The predictor variables were determined based on the daily climatic information and were clustered into two sets to capture two modeling patterns, namely, seasonal aggregation (Model A) and growth stage disaggregation (Model B). In Model A, the climate variables were summed up to include the total growing period of wheat (November 1 - April 30). The variables were; (i) mean maximum temperature (average daily maximum temperatures), (ii) mean minimum temperature (average daily minimum temperatures), (iii) total precipitation (sum of daily rainfall). These variables are the general conditions of thermal and moisture and are compatible with the conventional methods of crop and climatic modeling and thus form a point of reference. In Model B,

variables of climate were disaggregated using three broad phenological stages, including flowering (February 1 -15), grain filling (February 16 -March 15), and maturation (March 16 – April 10). The variables calculated at every stage included (i) mean maximum temperature (°C), (ii) mean minimum temperature (°C), (iii) total precipitation (mm). The 35°C temperature limit of extreme heat was established using physiological studies proving visible heat exhaustion in wheat especially at reproductive and grain-filling stages, which was the exact situation leading to the heatwave in 2022 (Porter and Gawith, 1999).Each of the three growth stages has had four variables calculated to obtain twelve stage-specific predictors and to also ensure consistency with the underlying dimension of the seasonal model of the climatic conditions. This design maintains the basic dimension of climatology and increases the temporal resolution by disaggregating phenological data. The fit of the two models was mainly compared through the goodness-of-fit power and a secondary F-test was used to establish whether the solutions of greater complexity of the Model B could generate a statistically significant boost in explanatory power.

3.4 Transformation of Yield and Model Specification.

Our dependent variable was yield anomalies per observation: this was the difference between the observed yield of that year and the mean of thirteen years of that district. This change plays a very important role. The range of the district-level mean yields was between 36.1 and 49.8 q/ ha (Table 1) which is an indicator of the variation in the productivity of baseline due to the differences of the soil quality, irrigation infrastructure, application rates of fertilizers and management practices- which are relatively unchanged in the thirteen years duration of our analysis. By modeling anomalies rather than absolute yields, we get our peak consideration on year-to-year climate-related variation and that is what our seasonal and growth stage climate characteristics should clarify. This procedure is typical to climate-yield analysis when operating across sites having varying baseline productivity. We used the ordinary least squares (OLS) regression on either of the models. Although there are more complex forms of machine learning, OLS regression is designed with three useful properties in the context of our research question: (1) easy interpretation of the effects of individual features through regression coefficients, (2) established statistical tests of the existence and comparison of models, and (3) easy comparison with published literature in fields of application of crop-climate models. The climate features were all modeled after being standardized (mean-centered and divided by standard deviation), putting them on similar scales and allowing the direct comparison of the magnitudes of the coefficients directly.

3.5 Framework of Statistical Comparison

3 The analytical structure was constructed as a controlled comparison to isolate the effect of temporal disaggregation. Two parallel regression equations were estimated Model A, which included three seasonal predictors, and Model B, which included twelve growth stage-specific predictors. The same dataset (169 observations of district-years) was used in the estimation of both models, but the two different models were identified with differences only in the level of temporal aggregation. Model performance was evaluated using standard goodness-of-fit metrics, including the coefficient of determination (R^2), adjusted R^2, root mean square error (RMSE; q/ha), and mean absolute error (MAE; q/ha). The adjusted R^2 was of specific interest considering that there are more predictors in Model B. To statistically determine whether the extra predictors in Model B offered a statistically significant enhancement in explanatory capability, an F-test for nesting models was carried out. This test is used to compare residual sum of squares (RSS) of the two models under the null hypothesis that the extra parameters of Model B do not improve model fit. The F-statistic is based on an F-distribution whose number of $df_1 = (p_2 - p_1)$ and $df_2 = (n - p_2 - 1)$ where n denotes the sample size and p is the number of predictors.This enables a rigorous assessment of the information provided by the disaggregation of time on top of the seasonal aggregation. Along with statistical significance, Cohen f^2 was also obtained as an effect size measure to determine practical significance regardless of sample size. In accordance with standard levels, f^2 values 0.02, 0.15, and 0.35 are associated with small, medium and large effects respectively. Pearson correlation analysis was also conducted between individual climate variables and yield anomalies to identify the possible feature-level relationships. To establish statistically significant relationships under the null hypothesis of no relationship, correlation coefficients and the correlational p-values were obtained. Such an analysis can help determine which phases of development and climatic variables have the most significant impact on the variability of wheat yield, and hence design specific adaptation methods.

3.6 Quality and Reproducibility of data

All data were processed in Python 3.9 using standard scientific libraries: pandas to manipulate data, numpy to operate numerically, scikit-learn for regression modeling, SciPy statistical tests, and matplotlib to visualize results. The total number of raw climate data was 62,140 daily observations across 13 districts over 13 years. We then filtered our climate data to the wheat growing season (November-April) and filtered out 0.3% of our records with missing or implausible values (usually
-999 flags of missing satellite retrievals) to get our processed climate dataset of 61,724 daily observations. Quality control involved checking whether values were physically implausible (values below -10°C, or higher than 50°C, more than 200mm/day precipitation), temporal consistency (e.g., no future timestamps), and whether there are discrepancies between the NASA POWER and ground station data (where ground station availability was available). The time series were visually inspected to confirm expected seasonal patterns and not suspicious artifacts. The yield data did not have any outliers that needed to be excluded; all district-year values fell within reasonable ranges given regional productivity and known climate events, including the documented low yields during the 2022 heatwave.

RESULTS

4.1 Descriptive Statistics

A total of 169 independent and district-year observations of 13 districts consisting of 13 consecutive harvest years (2010- 2022) form our last dataset. Table 1 includes features of yield in the form of a district level, and it is evident that there are significant differences in productivity in the form of the baseline. The mean yields were between 36.1 q/ha in Saharanpur and 49.8 q/ha in Bathinda which indicated the differences in the quality of soils, irrigation facilities and management activities in the area of study. Year-to-year variation averaged 3.4 q/ha (CV: 7.8%), indicating climate- driven fluctuations despite stable management. The 2022 harvest year, affected by the March heatwave, contributed notably to this variation with below-average yields across most districts.

4.2 Comparison of the performance of models.

The disaggregation of growth stage enhanced predictive performance by a very high margin as compared to seasonal averaging (Table 2, Figure 1). Model B accounted 19.9 % of variation in yields, which is almost fivefold of the 4.0 percent accounted by Model A. This proved highly statistically significant ($F(9,156)=3.43$, $p=0.0007$) with Cohen's $f^2=0.198$ which is medium in practical effect size. Both R2 and RMSE comparison between Model A and Model B plot in a bar chart]. In addition to overall fit, Model B also found substantially more statistically significant predictors. There were seven statistically significant ($p<0.05$) features of the growth stage compared to a single statistically significant ($p<0.05$) feature of seasonal, with various implications to climate adaptation.

TABLE 2: MODEL PERFORMANCE COMPARSION

Metric	Model A (Seasonal)	Model B (Growth Stage)	Improvement
R^2	0.040	0.199	+392%
Adjusted R^2	0.023	0.138	+500%
RMSE (q/ha)	4.09	3.74	–8.6%
MAE (q/ha)	3.16	2.88	–8.9%
Significant features ($p<0.05$)	1/3	7/12	+6 features
F-statistic	—	3.434	—
p-value	—	0.0007***	—

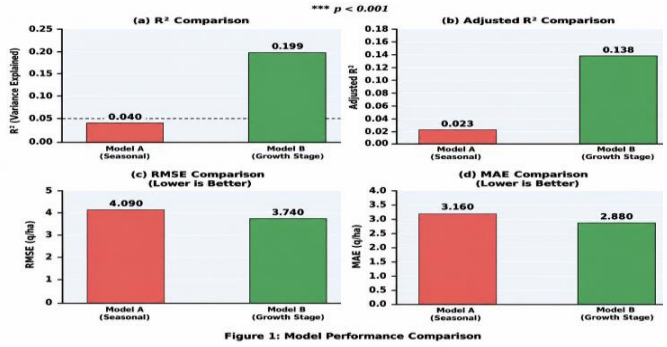

Figure 1: Model Performance Comparison

4.3 Level of Analysis

Correlation analysis revealed distinct patterns across growth stages (Table 3, Figure 2). The semi-arid study area we examined was not in agreement with theoretical predictions that include flowering sensitivity where the most significant predictor was found to be the climate during the maturation period. Table 3 presents the top climate predictors ranked by correlation strength.

The highest individual correlation was observed for maturation extreme heat days (r = 0.206, p = 0.007), indicating the presence of a strong negative impact of heat distress on the yields in the period of grain hardening (March 16–April 10). This observation is the direct reason behind the effects of the 2022 heatwave: very high temperatures in the process of maturation resulted in premature senescence and a decrease in grain weight. Surprisingly, maturation rainfall was negatively correlated with (r = -0.193, p = 0.012), and excess water can bring down harvest activities and make it more vulnerable to disease or pre-harvest sprouting.

The period of grain-filling was also critical to temperature. High temperature events (r = 0.191, p = 0.013) and mean level of maximum temperature between the periods of February 16 and March 15 (r = 0.186, p = 0.015) influenced significantly on yields. This period leads to faster development of the crops, this is caused by high temperature that will reduce the length of grain filling and results in low kernel weight at harvest. This heatwave of 2022 occurred exactly in this dangerously susceptible period, and it shows why critical heatwave monitoring is needed at the growth stage.

The flowering effects were also noteworthy but not as important (r = 0.168, p = 0.029) in comparison with subsequent stages of growth. However, physiological work focuses on heat sensitivity in flowering, but empirical findings of ours suggest that heat and moisture stress in late grain development processes could prove as much or more restrictive, which was evidenced drastically the 2022 event, in the semi-arid conditions of North India.

Out of all seasonal variables, minimum temperature was the only significant variable (r = 0.184, p = 0.017) indicating that the high growing season temperature was positively correlated with yields. This is probably a positive early-season growth, and seasonal averaging cannot separate early-season growth, with its likely effects of late-season stress, and seasonal growth.

TABLE 3: TOP CLIMATE PREDICTORS RANKED BY CORRELATION STRENGTH

Rank	Feature	Correlation (r)	p-value	Growth Stage	Climate Variable
1	Maturation extreme heat days	+0.206	0.007**	Maturation	Temperature extreme
2	Maturation rainfall total	-0.193	0.012*	Maturation	Precipitation
3	Grain filling extreme heat days	+0.191	0.013*	Grain filling	Temperature extreme
4	Grain filling Tmax mean	+0.186	0.015*	Grain filling	Temperature average
5	Flowering Tmax mean	+0.168	0.029*	Flowering	Temperature average
6	Grain filling day-night range	+0.154	0.049*	Grain filling	Temperature variability
7	Maturation Tmax mean	+0.145	0.067	Maturation	Temperature average
—	Seasonal Tmax mean (best seasonal)	+0.184	0.017*	Seasonal	Temperature average

4.4 Aggregated Growth Stage Contributions

To establish the contribution of each of the developmental stages to variation in yield, we compared the mean absolute correlation (|r|) between climate features in each growth stage (Figure 3). The results show that there are marked variations in the importance at each of the stages. The maturation stage turned out to have made the best mean correlation with the yield anomalies (mean| r | = 0.168), then grain filling (mean| r = 0.152), and flowering stage was comparably weak (mean| r |= 0.124). These findings show that variability in climatic conditions between the end of the season is the main determinant of the variability in yielding of wheat in the study area a finding that the catastrophic effects of the March 2022 late-season heatwave confirm.

Even though flowering is a statistically significant physiological event, it is not as important as the cumulative stress of grain filling and maturation. Conversely, climate aggregated over the growing season as a whole had weaker average relationships, which gives seasonal averaging its

weaknesses. Seasonal aggregation distorts stage-specific sensitivity to climate that are vital in yield formation by equalizing variability across time - specifically the reason behind why seasonal models could not anticipate the impacts of the 2022 heatwave in spite of the fact that the underlying daily data contained extreme conditions.

In general, it can be seen that yield formation is not limited to one crucial phase but rather many other development phases influence yield formation with maturation and grain filling having the greatest impacts on yield in a semi-arid environment in North India. This undermines traditional paradigms of wheat production which focus on the sensitivity of flowering stage and the fact that region-specific empirical validation of physiological response is necessary especially under conditions where there are heat and moisture stresses during the later seasons.

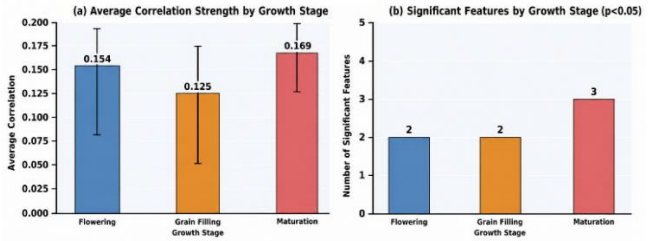

Figure 3: Growth Stage Climate Contributions to Yield Prediction

DISCUSSION AND CONCLUSION

5.1 Key Dependent Results and Implications

This experiment aimed at testing the hypothesis that disaggregating climate into growth stage-related aspects is better at predicting the yield of wheat over the traditional seasonal averaging. We discovered, using thirteen years of North Indian data that the growth stage features were able to explain almost 5 times the amount of yield variance than had seasonal averages (R2=0.199 vs 0.040, p<0.001). The seven variables were statistically significant compared to only one variable season, maturation and grain filling periods have become the most significant predictors. These findings support the assertion that the temporal disaggregation of climatic-yield relationships would be captured by seasonal averaging, which hides agronomically significant relationships between climatic variables and yields. This represents a 392% increase in explained variance. Despite the relatively low baseline of the seasonal model, this constitutes a substantial gain in information from the same underlying data. More to the point, the stage of growth analysis determines when climate stress is the most important, which would allow one to make specific rather than general adaptation suggestions.

5.1 Regional Background

The primacy of maturation period climate in our findings deserves special attention, as it diverges from the flowering- centric narrative common in wheat physiology literature. Three factors likely explain this regional pattern. First, semi- arid conditions in North India create particular vulnerability to late-season stress. By March–April, temperatures routinely exceed 35°C while soil moisture declines, creating heat– drought combinations that accelerate senescence and reduce grain weight. The March 2022 heatwave exemplified this vulnerability: temperatures exceeding 40°C during late February and March coincided precisely with grain filling and early maturation, causing yield reductions of 10–15% across the region. What made this event particularly instructive was that seasonal climate averages for the 2021–22 growing season remained within historical norms. Had seasonal models been used for forecasting in early 2022, near-normal average temperatures would have suggested typical yields. In contrast, a growth stage-based approach, tracking extreme heat days during specific developmental windows, would have flagged elevated risk in advance—demonstrating not only statistical superiority but also operational value for climate risk management. Second, our measure—extreme heat days during maturation—likely captures compound stress more effectively than mean temperature alone. Third, the negative relationship between maturation rainfall and yield suggests that in this region, excess late-season moisture (although uncommon)

can disrupt harvest operations and reduce grain quality. This regional specificity, validated by the 2022 natural experiment, highlights a key advantage of empirical approaches. They reveal location-specific climate sensitivities that may diverge from global physiological patterns, thereby providing evidence-based guidance for adaptation planning.

5.2 Comparison With Published Studies

The growth stage model ($R^2 = 0.199$) is lower than the $R^2 = 0.25$– 0.35 typically reported in large-scale seasonal analyses (Ray et al., 2015; Zhao et al., 2017). However, direct comparison presents several challenges. First, our analysis is conducted at the district level, where greater unexplained variability arises from within-district heterogeneity in soils, management practices, and microclimates—factors that tend to average out in large global datasets such as the 13,500 observations used by Ray et al. Second, non-climatic variables (e.g., irrigation, fertilizer use, and crop varieties) were intentionally excluded to isolate climate effects, although their inclusion would likely improve explanatory power. Third, the relatively small sample size (n = 169) limits the achievable R^2 compared to large datasets. Importantly, the key contribution of this study lies not in absolute R^2 values but in relative improvement under controlled conditions. The growth stage model explains more than four times the variance captured by seasonal features using the same dataset. This controlled comparison—unlike cross-study contrasts—demonstrates unequivocally that temporal disaggregation provides less ambiguous and more informative insights into climate–yield relationships.

5.3 Policy Recommendations to Adapt to a Changing Climate.

Findings in growth stage also allow specific adaptation strategies. The realization that the extreme heat of maturation is the most potent variable forecasting yield losses implies three possible measures, namely: (1) the cultivation of early-maturing varieties that would fill the grain before the extreme heat in April hits; (2) the manipulation of the planting date to achieve a later maturation date of March over April; and, (3) a focus on the allocation of irrigation to soothe late-season moisture stress. These particular recommendations are in opposition to generic advice such as to raise heat tolerance, and this goes to show how practical the concept of temporal disaggregation is. In a similar vein, the grain filling temperature impact has a case in favor of varieties that fill their grain rapidly and therefore spend minimal time in exposure and the impact of flowering temperature is vindicated in independent attempts to utilize heat-tolerant reproductive phenotyping. A seasonal solution would merely suggest the drought and heat tolerance but leave when and how to act to be left out.

5.4 Limitations and Future directions.

Three major limitations that should be mentioned are discussed. The calendar-based developmental stages used provide only an estimate of the true crop phenology, which is dependent upon the dates that are used to plant and the choice of cultivar. This may be filled by future studies by adding the phenological models which dynamically vary the stage timing in response to year and place-specific data, especially where there would be information available on the sowing dates and varietal selection. Second, the aggregation employed on a district level in this study can underscore both within-district heterogeneity; with farm-level research one would be better able to estimate than control farm- level climate effects by managing heterogeneity. Third, the sample size is rather small (thirteen years in a single area) and does not indicate the generalizability of the study results, which necessitates expanding spatial and time coverage. The extent of their practical implications was made plain during the heatwave incident of 2022. With growth stage monitoring systems in place, the predicted drastic heat in late February- March would have been used to implement specific measures, such as emergency irrigation at the farm level during grain filling, early harvest advice where possible, and district level compensation strategies. The effectiveness of these reactions also requires prediction of the occurrence of heat stress but also prediction of when it happens concerning crop growth; a phenomenon that growth stage-specific methods alone can provide. Future research should thus look into whether the noted predominance of maturation-stage climate influences is universal to all other semi-arid wheat-producing areas or the North India area. Comparison of agro-climatic zones would aid in verifying whether a growth stage disaggregation, which is universally defined or that of region specific is more predictive. Moreover, combining process-based modeling of crops to include growth stage sensitivity to climatic sensitivity has brought forth an alternative channel of filling the gap between statistical and mechanistic models, which could improve physiological realism and forecasting abilities.

5.6 Conclusion

Crop-climate modeling has also been successful using seasonal climate aggregation, and strong correlations have been determined between the weather and yields around the world. But homogenising whole growing seasons is at odds with physiological evidence of stage-specific climate sensitivity and leads to the loss of playing field provided by the time series information within daily climate data. The controlled comparison in this study proves that disaggregation at growth stage significantly enhances capability of prediction of yield besides the determination of certain critical periods. In the case of North Indian wheat, climate combining maturation and grain filling was found to be more predictive compared to flowering conditions, which defied the individualistic assumptions of physiology of temperate regions. Such a local peculiarity emphasizes the importance of both empirical validation and theoretical knowledge. In a more general sense, our results indicate that future generations of crop-climate models must regularly consider growth stage specificity, especially as climate change elevates the frequency of extreme events during sensitive developmental periods. The methodological framework that we present blueprints on how similar studies can be done in other crops and other geographical areas, the operational impact of which will enhance the science knowledge base and also the practical implementation of climate adjustment in agricultural systems across the globe.

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